Tanya Singh

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EDUCATION

University of Maryland, College Park **College Park, MD** **Graduation Date: May 2027**

Bachelor of Science in Computer Science, Minor in General Business *GPA: 3.985, Dean's List*

- Relevant Coursework: Object-Oriented Programming, Applied Probability and Statistics, Discrete Structures, Computer Systems, Linear Algebra, Organization of Programming Languages, Algorithms, Data Science, Advanced Data Structures, Machine Learning, Compilers
- Leadership: Committee Member at Association for Women in Computing, Marketing Lead at Google Developer Student Club, Committee Head at Campus Coders Crew, Researcher in the First-Year Innovation & Research Experience

SKILLS

- **Languages:** Python, Java, C, SQL, R, OCaml, Rust, JavaScript, HTML/CSS, MATLAB, x86 Assembly, Haskell, Racket, Dafny
- **Libraries:** Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Tools & Platforms:** Microsoft 365 Copilot Agent Builder, Git/GitHub, Azure DevOps, ServiceNow, Microsoft Excel, VS Code, RStudio
- **Concepts:** Machine Learning (Classification, Clustering, Regression), Large Language Models, Prompt Engineering, AI Agent Development, Knowledge Base Design, Hypothesis Testing, Data Visualization, Data Analysis, Agile/Scrum, Zero Trust
- **Certifications:** Microsoft Office Specialist (Associate)

WORK EXPERIENCE

Navy Federal Credit Union **Vienna, VA** **May 2026 - August 2026**

Summer Associate

- Deployed an M365 Copilot-powered Zero Trust AI Assistant that provides cited answers on 29 active projects, 45 capabilities, and 152 activities.
- Designed a metadata-driven knowledge base by cataloging 84 Zero Trust documents and standardizing AI retrieval across teams and workstreams.
- Reduced AI response times by 90%+ by cutting retrieval latency from 2+ minutes to 10 seconds through prompt engineering and optimization.

Lavner Education **Washington, DC** **June 2025 - August 2025**

Intern/Instructor

- Instructed 120+ students in coding, STEM & Medicine, All-Girls STEAM, robotics, and more through interactive, age-appropriate lessons.
- Mastered a hands-on curriculum to deliver a one-on-one learning experience to 40+ students across a wide variety of STEM subjects.
- Led set-up and maintenance of on-site check-in and computer software, increasing work efficiency by 20%.

Easy Dynamics **McLean, VA** **January 2025 - January 2025**

Intern

- Evaluated the ability of 4+ LLMs to help researchers find, collect, and summarize publicly available information on payroll threats and fraud.
- Designed 12+ research prompts to compare responses, analyzing each LLM's strengths and weaknesses on accuracy and efficiency.
- Developed and presented a proposal on findings and recommendations for using LLMs in this research, built with PowerPoint, Excel, and Word.

Code Ninjas **Leesburg, VA** **November 2022 - November 2023**

Code Sensei

- Taught 50+ students aged 5 - 14 to write and understand code in programming languages such as Scratch, JavaScript, C#, and more.
- Applied the center's curriculum to guide students through 5+ software platforms such as Roblox Studio, MCreator, and Unity.

PROJECTS

Machine Learning Stroke Risk Prediction | *Python* **2026**

- Developed an end-to-end stroke prediction pipeline in Python using Pandas, NumPy, and Scikit-learn on 5,110 patient records. Applied data cleaning/preprocessing, feature engineering, and feature scaling to compare Random Forest and Logistic Regression models, analyzing age, glucose, and BMI as prominent features.

Lexer, Parser, and Interpreter | *OCaml* **2025**

- Engineered an OCaml-like dynamically typed programming language by building a full lexer, recursive-descent parser, and interpreter. Applied language design principles, grammar-based parsing, lexical scoping, closure formation, and environment-based evaluation, building a stronger understanding of programming language implementation.

Performance Optimization & Multithreading | *C* **2025**

- Designed and implemented a custom dynamic memory allocator in C to explore explicit memory management and optimization. Applied multithreading with pthreads, mutexes, and barriers to build a matrix normalizer, addressing concurrency, synchronization, and cache efficiency in parallel processing.

Personal Portfolio | *HTML/CSS, JavaScript* **2025**

- Created a fully responsive and interactive personal website from scratch using HTML/CSS and JavaScript, deployed on a custom domain. The website features smooth animations, a filterable projects page, and separate pages for experience, research, coursework, and contact to showcase professional experiences.

RESEARCH

Effectiveness of Large Language Models (LLMs) in Weather Forecasting for Arlington, VA | *R* **2025**

- Co-authored a research paper evaluating Large Language Models (GPT-5, Claude Sonnet 4, and Gemini 2.5 Pro) to forecast 92 days of Arlington, VA weather from one year of historical data. Applied prompt engineering, cleaned and compiled model outputs, and ran paired t-tests and Wilcoxon signed-rank tests in R. All three models differed significantly on high temperatures ($p < 0.001$), while GPT-5's low-temperature predictions showed no statistically significant difference from observed temperatures ($p = 0.66$).